

Neuromorphic Hardware Solves Complex Math Efficiently

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Brainlike Computers Can Do Math, Too

Neuromorphic computer solves differential equations

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Brad Theilman, a computational neuroscientist at Sandia National Laboratories, helped discover that nature-inspired, neuromorphic computers, like the one shown here, are better at solving complex math problems than previously thought.

Craig Fritz

Computer scientists often assume that the brain works by approximations, and therefore that computing hardware inspired by the brain won't be as good at complex math as traditional hardware. Researchers at [Sandia National Laboratories](#), in Albuquerque, are pushing back against this premise. In a paper published in [Nature Machine Intelligence](#) last November, they show that neuromorphic hardware built by [Intel](#) can solve differential equations using one of the most important methods in [scientific computing](#): the finite element method.

Computing inspired by the brain

[Neuromorphic computing](#) promises to be more energy efficient than conventional hardware, which uses densely packed electrical switches to add up signals. "We have made tremendous advances in AI, but people are building power plants" to make that possible, says [Sandia](#) computational neuroscientist [James B. \(Brad\) Aimone](#). "Meanwhile, we're able to have this conversation at 10 watts each," he says, referring to the power consumption of the [human brain](#).

The brain relies on relatively sparse [neurons](#) that communicate with pulses in spiking patterns. "Neurons receive weighted information and send a timed, all-or-nothing pulse that is transmitted to near neighbors," Aimone says. "The dynamics of this determine the output." There are various ways of implementing these spiking patterns in hardware. Some systems use analog elements; others, like Intel's [Loihi 2](#), replicate these spiking neurons in digital circuits built with conventional manufacturing methods. Loihi 2 has over 1 billion of these digital neurons, a similar number to those found in the brains of small mammals and birds.

Typically, people working in [neuromorphic computing](#) focus on applications they assume are brainlike, such as processing real-time sensor data, says Aimone, whose background is in [neuroscience](#). He says this may be underestimating the capabilities of the brain, and that there's no reason to assume this hardware isn't suited to conventional [high-performance computing](#) tasks like the finite element method.

The finite element method, or FEM, is widely used to solve problems in fluid dynamics, mechanics, and electromagnetics—how materials fail, how [Wi-Fi](#) signals travel through a building. It's a way to solve differential equations related to "any physical problem that's distributed over time and space," says [Bradley Theilman](#), a computational neuroscientist at Sandia.

The inspiring monkey brain

Theilman and Aimone assert that the brain routinely solves similar problems. Hitting a [baseball](#), for instance, is a complex physical problem that requires processing information that's changing over time—following the arc of the ball and planning how to move the bat. “It’s a complex problem. The brain is controlling muscles in response to real-time information to make contact with the ball,” Theilman says.

Their research on the finite element method was inspired by a computational model of the motor cortex in a monkey. A table of numbers in this model, called a matrix, reminded Theilman of matrices used in the finite element method. The Sandia team translated the FEM to this motor cortex model, implemented it on Loihi 2, and showed that it can solve partial differential equations.



Brad Theilman [center] and Felix Wang [in [deep blue](#) hoodie] unpack a neuromorphic computing core at Sandia National Laboratories. Craig Fritz

Theilman says their projections suggest that doing FEM on neuromorphic hardware may offer a slight energy advantage over traditional systems, but since they are using non-standardized research hardware it's challenging to make a realistic comparison based on the results they've published so far. The team is currently working on adapting other larger problems to the neuromorphic hardware, and they expect there may be clearer energy advantages on larger scales. Standard computational methods have been optimized for traditional hardware, and vice versa, says Aimone. It will take time to show [energy efficiency](#) advantages.

[Steve Furber](#), a computer scientist emeritus at the University of Manchester, in England, says this builds on the Sandia team's [previous work](#) implementing another method for solving differential equations, the [Monte Carlo](#) method, on neuromorphic hardware. He says Loihi 2 is well suited for solving these sorts of problems, and should be efficient at them. Furber's team has used its Arm-based [SpiNNaker](#) hardware for modeling heat diffusion, which is a similar problem.

Aimone says these advances show that neuromorphic computing has broader potential than computer scientists have appreciated. "It's worth looking deeply at any kind of mathematical problem," he says. "There's no reason to assume you can't do something in neuromorphic computing."

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